

Control a mobile robot in Social environments by considering human as a moving obstacle

S. D. N. Tanha

Robotics Research Laboratory, School of
Mechanical Engineering
Iran University of Science and
Technology
Tehran, Iran

S. F. Dehkordi*

Robotics Research Laboratory, School of
Mechanical Engineering
Iran University of Science and
Technology
Tehran, Iran
s_fathollahidehkordi@mecheng.iust.ac.ir

A. H. Korayem

Robotics Research Laboratory,
School of Mechanical Engineering
Iran University of Science
and Technology
Tehran, Iran
hkorayem@gmail.com

Abstract— in this paper a comprehensive control method for a mobile robot with non-holonomic constraints, which can be used in dynamic environments with fixed and moving unpredictable obstacles presented. This issue becomes more complicated when obstacles involve humans act as a moving obstacle without any data while erratic behaviors and reactions are important. In this regard, fuzzy logic algorithm whose rules are developed for these conditions suggest for two distinct strategies: (1) Regulation between two points and (2) Path tracking. The main contribution of this paper includes defining the fuzzy logic role for human environments which are affected the robot speed and rout, considering the target position and its variation during the robot motion. The proposed controller algorithm simulated in three modes: (1) Fixed obstacles, (2) Moving obstacles and (3) Moving both of obstacles and target. These approaches can be used in social robotic, humanoid robots, and educational robots.

Keywords—mobile robot, social robot, fuzzy logic, moving obstacle, moving target.

I. INTRODUCTION

One of the applications that can be considered for a mobile robot is the use of robots as a social interface in public places such as shops, airports, etc. Mobile robots are subject to restrictions in relation to moving in different environments, as well as the need to identify the work space around them. These limitations include the presence of obstacles, the path of robot, speeds of movement, and so on. Due to the growing use of robots in environments where humans are present, it is necessary to update the behavior and plan of this kind of robots such as social or human partner robots. In this study, we focus on control a mobile robot in environments where humans are present in addition to other fixed and moving elements. In Figure 1, schematic of the work space of a social robot has been shown. The purpose of this social robot is to follow the path and reach the person who calling the robot. In addition to providing a method to prevent collision with the obstacles that shown in this image as individuals present in the environment, the logic of robot movement is extracted so that the presence of the robot in the environment does not affect the behavior of people. Due to the need for rapid decision making as well as planning in different situations, a fuzzy control algorithm is used to control the mobile base.

Korayem et al. proposed a control algorithm for mobile robot by using optimal control technique in the presence of moving obstacles. The path generation in this research was such that the robot were prevented from colliding with the obstacle [1]. Lee et al. have used the output feedback approach to control the motion of a mobile robot with non-holonomic and holonomic constraints [2]. Iqbal use a I/O linearization

feedback method to control the motion of a robot with two-differential driven wheel while a small Passive wheel considered. They are assuming that the environment is without obstacles [3]. Korayem et al. have devised mathematical modeling and optimal path generation for a mobile robot with a manipulator contains flexible link and joints, which uses open loop optimal control to track the desired path [4]. Wai has performed a non-holonomic mobile robot control using fuzzy neural network method [5]. Korayem et al. have implement and simulate the control strategy for a mobile robot by using optimal feedback control method [6]. Tipsuwan has developed a fuzzy logic and genetic algorithm for controlling the autonomous mobile robot [7]. Ghariblu and korayem present optimized trajectory for mobile flexible manipulator. Also, Korayem et al. performed path planning of a mobile manipulator based on the movement of the manipulator [11].

In this paper, we examine the problem and its conditions, the limits of the social robot as well as the logic adopted. Then, the simulation results are analyzed and eventually, the proposed method is discussed and some suggestions are made to continue and improve this research.

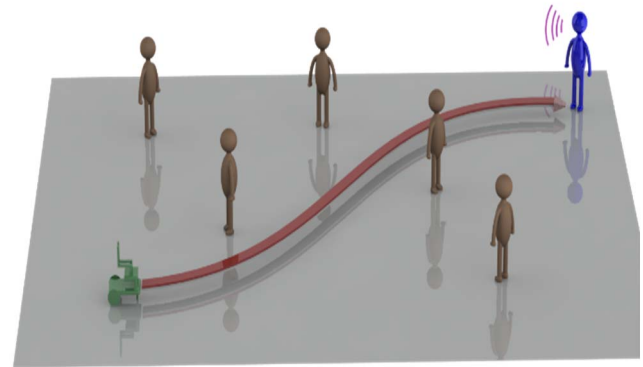


Fig. 1. Schematic design of the workspace of a social robot

II. FUZZY CONTROL OF SOCIAL MOBILE ROBOT

Generally, control of robots performed in two parts of path planning and path tracking. But by the use of fuzzy logic due to the nature of the stability of this type of controller, it is possible to route and track the path simultaneously while the control logic has been considered. In order to identify the inputs and outputs required by the control of the social mobile robot, as well as selecting the appropriate control algorithm, we first introduce the problem conditions.

A. The social environment

Controller of robot is planned in such a way that it can be used in a shop environment. Some of the features of a shop center as a good example of a social environment that can be mentioned is people's traffic, the possible passage of a freight carriage, the presence of columns and walls with fixed positions, etc., which are fixed and moving obstacles in the environment and can be considered in terms of their dimensions. In such an environment, the movement of obstacles is assumed unpredictable and the observance of the privacy of people is necessary. In Fig. 2, the schematic of privacy of people when the robot passes through its path is shown with a dashed line. The safe distance is proportional to the dimensions and speed of the robot. This radius defines the range that robot does not have the right to enter. Also, the boundary that shown with $R_{Limited}$ indicates that robot is required to slow down when it enters.

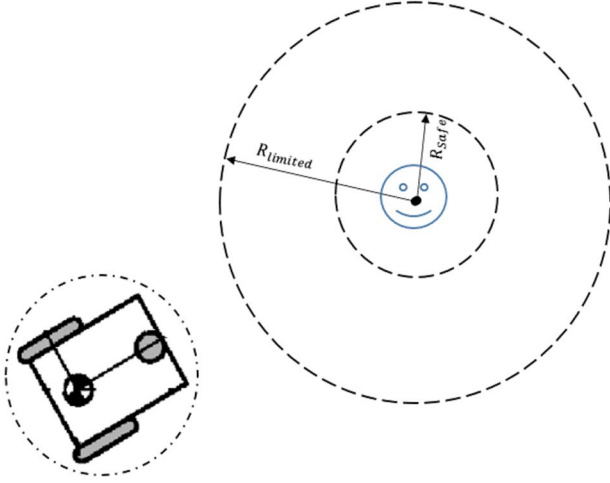


Fig. 2. Obstacle privacy and Boundaries

B. Controller Design

In this paper, robotic path planning is based on the momentary information received from the sensors. So that we cannot set the predetermined path for the robot to move. In such situations, the use of high-precision control algorithms is not possible to track the path. Because, the path is time-variable and movement of obstacles is completely random and non-modeling. It is also assumed that the top view of the environment that the robot moves is not available and it is not possible to specify the path from origin to destination without crossing the path. While in other path planning methods, to determine the path of the robot, information from the environment such as the above view needed to determine the position of the target, robot and obstacles.

1) *The control algorithm:* each fuzzy system has three main phases, fuzzifier, a decision making machine and a defuzzifier. To use fuzzy logic, it is necessary first to pass the inputs of a fuzzifier mechanism. In this study, a list of different modes that may be happened is created, and for each mode, the fuzzy parameter values are specified. Then, in decision making machine, for each of the predicted modes, a solution is chosen according to the conditions. Finally, with the defuzzifier, the path and the desired kinematic parameters

are created. Further, with solving the reverse kinematic of the robot, required control commands are created (Fig4.).

2) *Kinematics:* in this section, kinematics of a mobile robot with two differential wheels and a spineless wheel and the inverse kinematic equations is introduced in the state space. Consider the robot in figure 3 that can move in the XY Cartesian system. Angle between axis that Attached to the robot X_R and the axis X of the reference coordinate system X_I , is called head angle (θ). Also, the angle between X_R and the vector from center of mass of the robot to target position called α . Due to the nature of the two-wheeled mobile robot with two differential wheel, the direction of linear velocity of robot is same as X_R .

Therefore, linear and rotational velocities of robot are as follows:

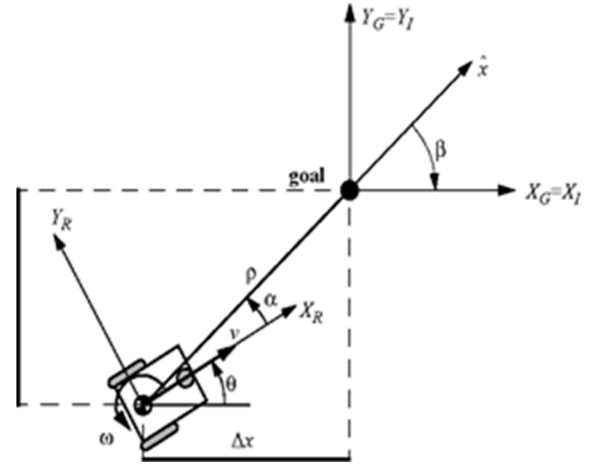


Fig. 3. Schematic of mobile robot with two differential wheels and target position in inertia system [8]

$$v = r \frac{\omega_1 + \omega_2}{2} = V_{XI} \cos(\theta) + V_{YI} \sin(\theta) \quad (1)$$

$$\delta\theta = \mu \frac{\omega_1 - \omega_2}{2} = \omega \delta t \quad (2)$$

In Equations (1) and (2), the values of ω_1 and ω_2 is rotational velocity of the robot wheels, r is the wheel radius, and μ is a coefficient to turn robot wheel's rotation into the base rotation around the center of the rotation, which depends on the geometric dimensions of the robot and wheels. Another parameter considered as input of the fuzzy controller is the rate that robot and obstacle approaches each other. Calculating and applying the effect of this parameter on the fuzzy controller, prevents robot's sudden and rapid movements to the obstacles and guaranteed the safety of the people in the robot's work space from being confronted with people. The value of this parameter is obtained from the following equation:

$$\Omega_i = |v - v_o| \quad (3)$$

Calculating Ω_i in polar system is more comfortable than Scular system . so :

$$\Omega_i = |dp_o + \rho_o d\alpha_o| = |dp_o| \quad (4)$$

That $|dp_o|$ is calculated with fuzzy logic of Table I in two consecutive time step .

Figure 4 shows status of ultrasonic sensores for diferent Obstacle Positions. With a fuzzy logic , α (angle betwin obstacle and robot) and ρ (distance betwin obstacle and robot) is determined .

3) Decision making: another fuzzy logic have been made for determining outputs of robot's controler. That controler used position of robot and obstacle and target as input and with a decision making kernell ,outputs (rotational speed of DC motors) determined. Decision making center based on avoidin obstacle colision and approaching target. The main

idea is choose betwin left and right or forward moving like a human. So table of states chooses as much as posible similar too human actions in same situations. in the next section , results of simulation of generated algorithm is peresented.

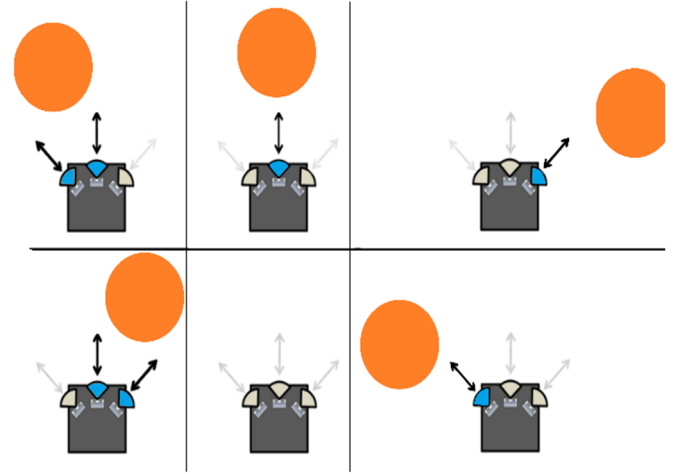


Fig 4. Status of ultrasonic sensores for diferent Obstacle Positions.

TABLE I. FUZZY LOGIC FOR DETERMINING OBSTACLE POSITION

If ... Sensor is			Then Obstacle Position is	
Left	Front	Right	α_o	ρ_o
S	S	F	L	S
S	M	F	VL	S
S	F	F	VL	S
M	S	F	Z	S
M	M	F	L	M
M	F	F	VL	M
F	S	S	R	S
F	S	M	Z	S
F	S	F	Z	S
F	M	S	VR	S
F	M	M	R	M
F	M	F	Z	M
F	F	S	VR	S
F	F	M	VR	M
F	F	F	No Obstacle	No Obstacle

III. SIMULATION RESULTS

Consider the Scott robot in Fig. 3, whose geometric parameters are defined in Table 1. The kinematic equations of motion of a robot with three degrees of freedom and the equation of motion of obstacles and the target with two degrees of freedom are simulated in the XY plane. In this simulation, the position of the obstacles and the goal is assumed to be fixed and moving in two ways. Also, a number of emotional parameters such as safe distance, speed of approach to the target, speed reduction near obstacles, maximum speed of the robot, and ... with repetition of simulation in different modes and empirically selected.

In Fig. 5, we see the simulation of the controller designed for a situation in which assumed that position of target and four obstacles are fixed. Also, simulating the path of the robot's movement in moving obstacles - fixed target and moving obstacles - moving target respectively in Figures 6 and 7 are shown. Figure 8 shows the linear speed of the robot in three simulated modes. It can be seen that its speed is reduced in proportion to the path that the robot enters the Limited Zone. In all three modes, the robot does not enter the Safe Zone anywhere.

It could be observed that in four steps the robot's speed is reduced by about 40%. Three times this speed falls in the vicinity of the obstacles and the final speed is reduced to approaching the target. In Table 2, you will see the least distant of robot with obstacles in the three simulated states.

TABLE II. SCOUT ROBOT SPECIFICATIONS, BASE PARAMETERS [11].

Description	Parameters	Value	Units
Wheels radius	r	0.08	M
Half of robot's width	b	0.145	M
Mass of base	m_b	6	kg
Mass of wheel	m_ω	0.32	Kg
Moment of inertia of base	I_{zzb}	0.06363	Kg m^2
Moment of inertia of wheel around rotation axis	$I_{yy\omega}$	0.00006	Kg m^2
Moment of inertia of wheel around axis liad on its diameter	I_{zzb}	0.00081	Kg m^2
Stall torque of motors	$u_{stall.r} u_{stall.l}$	6.85	Nm
No-load speed of motors	$\omega_{nl.r} \omega_{nl.l}$	14	rad/s
Control point	ω_s	0.1	m

TABLE III. MINIMUM DISTANCE BETWEEN ROBOT AND OBSTACLES IN 3 SIMULATION

Minimum Distance with (cm)	Sim 1 (Fixed Target)	Sim 2 (Moving Obstacle - Fixed Target)	Sim 3 (Moving Obstacle - Moving Target)
Obstacle 1	61	50	55
Obstacle 2	52	4	
Obstacle 3	97	9	
Obstacle 4	47	4	

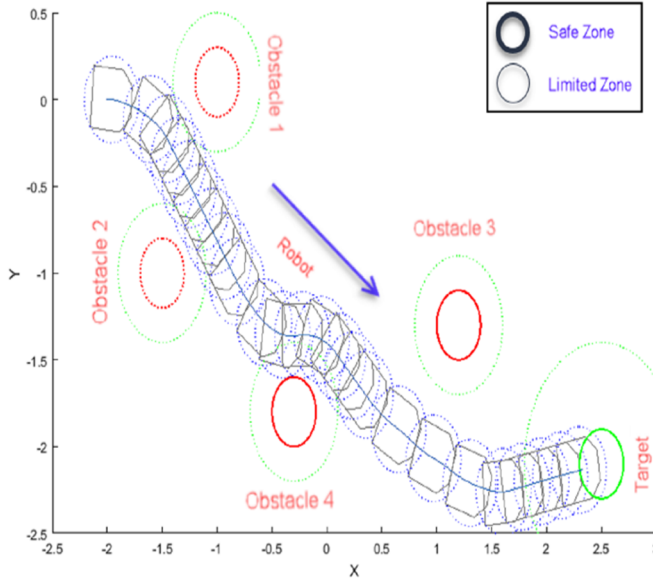


Fig. 5 Moving the robot toward the fixed target in the presence of fixed obstacles

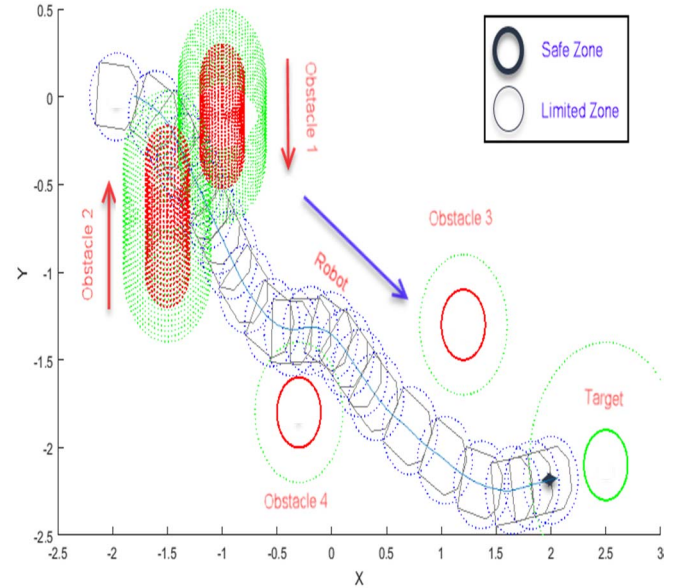


Fig. 6 Moving the robot toward the fixed target in the presence of fixed and moving obstacles

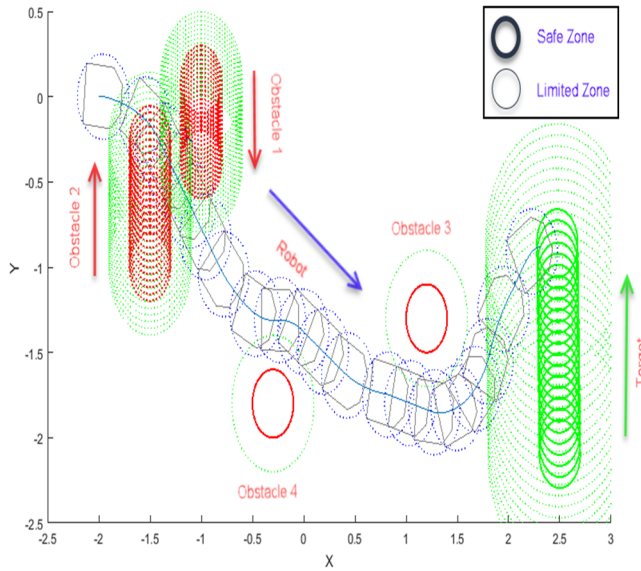


Fig. 7 Moving the robot towards the moving target in the presence of fixed and moving obstacles

IV. CONCLUSIONS

Robot control algorithm for moving between the two workspaces was presented taking into account conditions such as human presence in the social environment, randomize of obstacle movements, and following the moving target. Generally, the control of mobile robots and the path planning process are done in two ways, model base and non-model base. In this paper, the use of fuzzy logic in non-stationary simplified conditions was proposed for modeling mobile robot navigation in a social environment. The difference between the proposed algorithm and the model-base controller is the integrity of the robot's routing and control process. This means that the control commands are generated directly from the controller inputs and the path is not defined for the robot. The reason for this choice is the conditions governing the social environment and the randomistic nature of the movement of obstacles. The low volume of computations as well as the similarity of robot reactions to the human decision making model can be seen from other features of fuzzy logic as compared to model base controllers.

The control and guidance of mobile robots in the social environments have essentials that we tried to consider them in proposed fuzzy logic. It was observed that the speed of the robot in the limited zone was reduced by 40%. It also prevents to approach obstacles more than the Safe Zone that are considered in the simulations of the circle surrounding the obstacles.

In order to improve the performance of this controller, it is possible to use the genetic algorithm to optimize the path and reduce the energy consumption of the robot. Also, due to the modularity of the algorithm, each component of the algorithm can be updated or replaced by a more efficient method.

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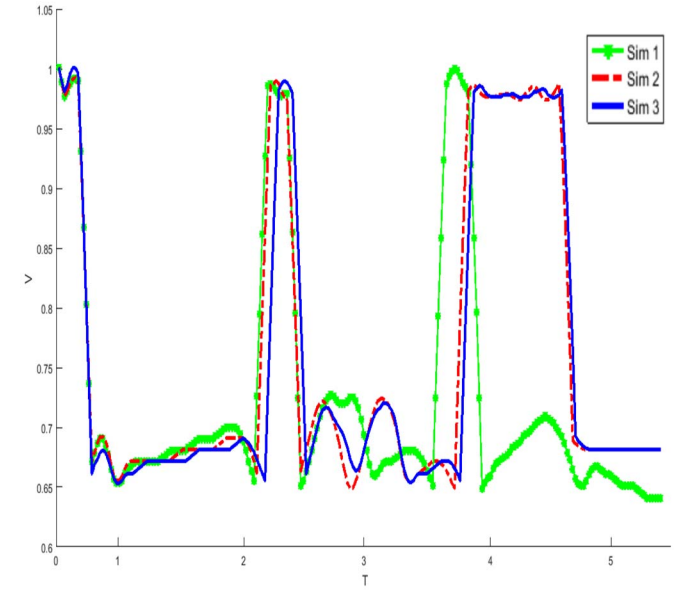


Fig. 8 Robot linear speed in three simulated modes

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